

Lab Task 05

Drawing Logical DFDs

1. Context-Level DFD of the Proposed System (Identify sources and sinks)

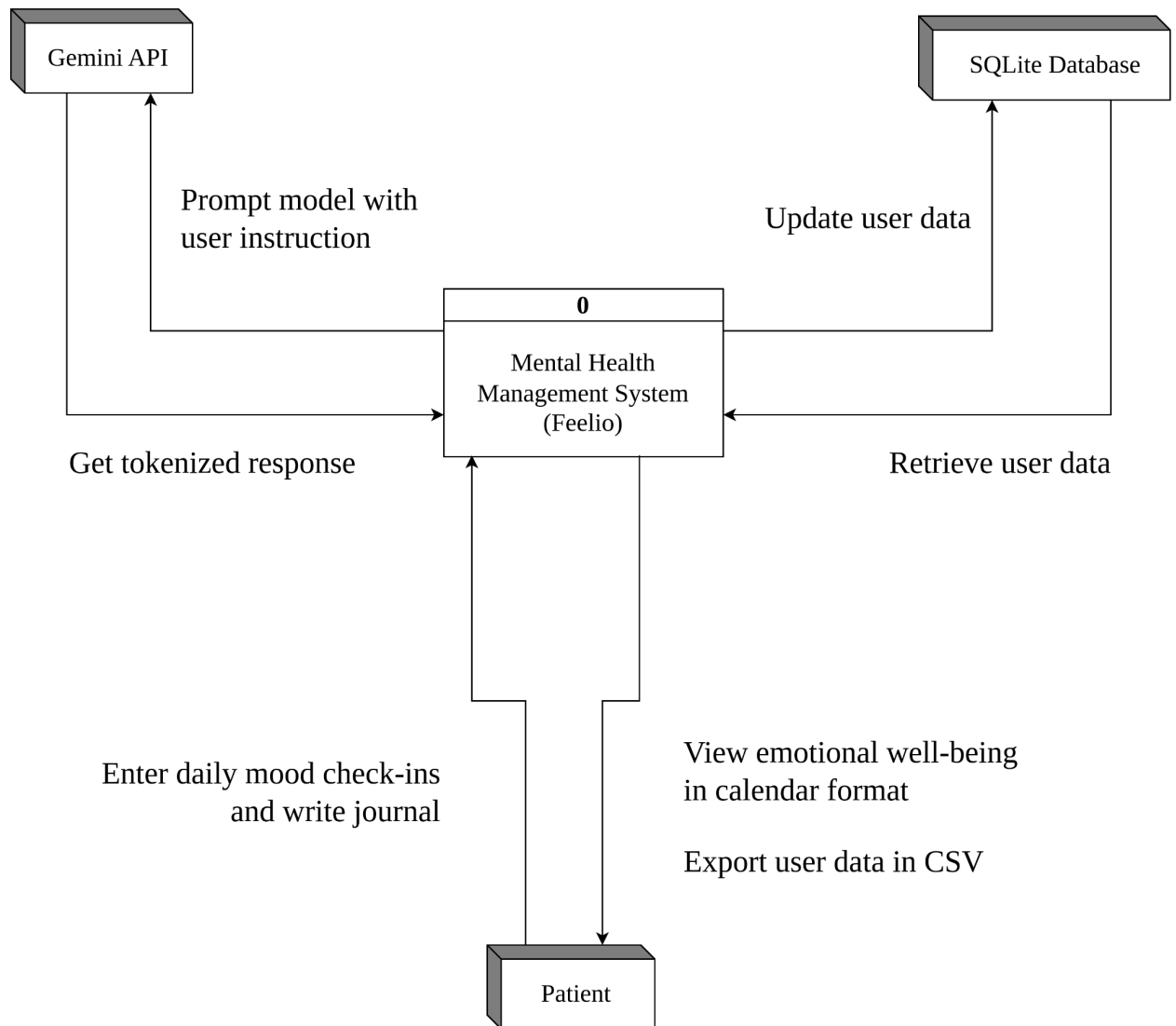


Figure: Context Level (Level 0) Data Flow Diagram.

2. Level 1 DFD

- a. Create a list of activities (External Entities, Processes, Data stores and Data flows)

External Entities:

- Gemini API
- SQLite Database
- Patient

Processes:

1. HTTP Fetch Service
2. Mood Tracking Service
3. Data Export Service

Data Stores:

- [D1] User Data

- b. Construct Level 1 DFD / Diagram 0 (Identify manageable subprocesses and major data stores)

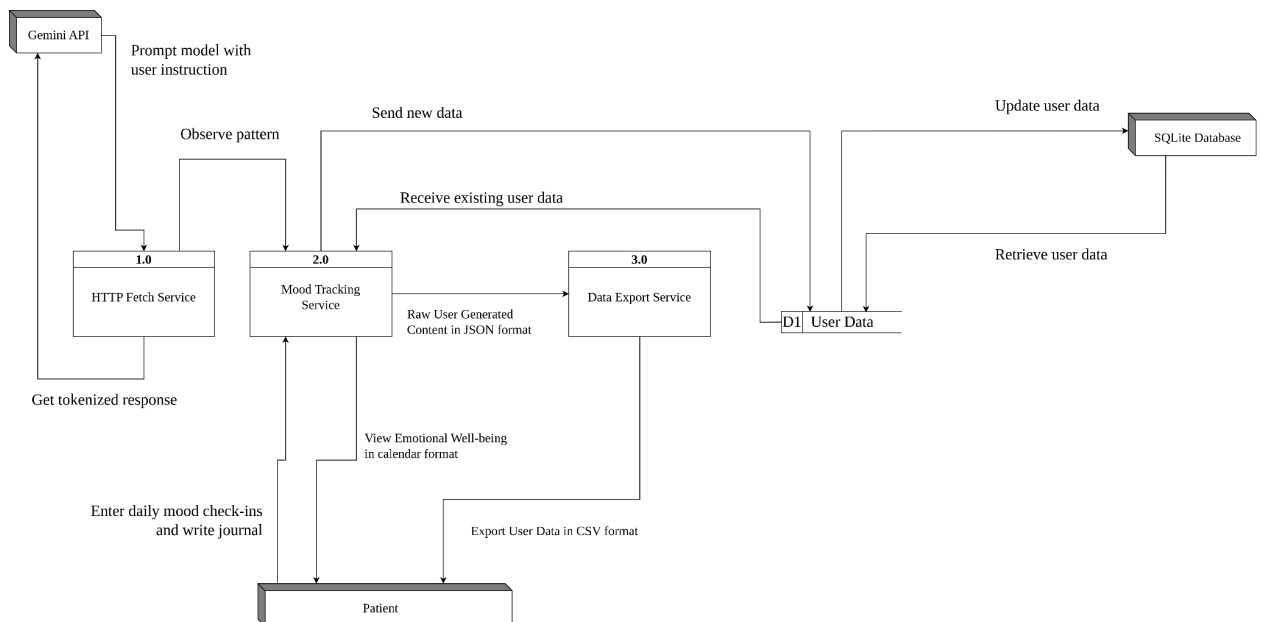


Figure: Level 1 Data Flow Diagram.

3. Level 2 DFD

- a. Create a list of activities (External Entities, Processes, Data stores and Data flows)

External Entities:

- Gemini API
- SQLite Database
- Patient

Processes:

- 1.1. Recursive Request/Response
- 1.2. Pattern Recognition Service
- 2.1. Calendar Service
- 2.2. AI Chat

- b. Construct Level 2 DFD / Diagram 0 (Identify manageable subprocesses and major data stores)

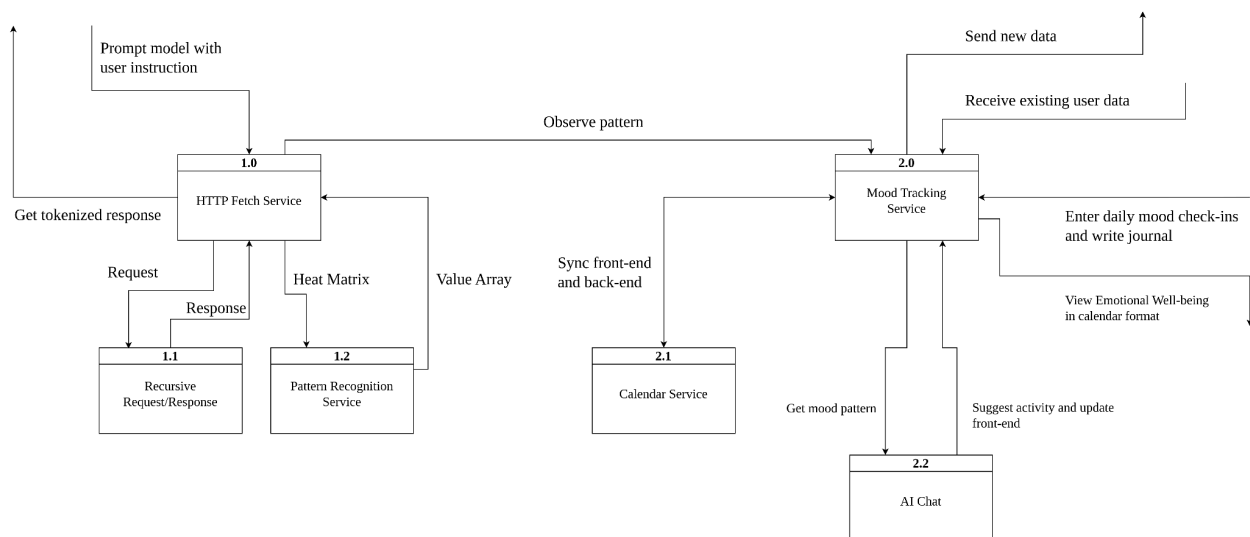


Figure: Level 2 Data Flow Diagram.

4. Construct the CRUD Matrix.

Function Table	Profile Information	Mood Check-In	Journal Text	Chat History	Recommended Content
Add User	CR	-	-	-	-
Update Info	U	-	-	-	-
Input Score	-	CU	-	-	-
Input/Edit Text Content	-	-	CRUD	-	-
New Message	R	R	-	-	CRU
Delete History	-	-	-	D	U
Visualize Calendar Score	-	R	-	-	-
Export	R	R	R	R	R